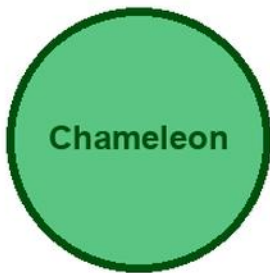


KITABU QUEST

Form II

BIOLOGY



Chameleon

Cover scene

students observing living things safely with a chameleon mascot

● Clear lessons

● Worked examples

● Fun activities

● Real-life problems

GRADE

II

Title Page

KITABU QUEST Form II Biology

Tanzania TIE-BASIC learner book

Mascot: chameleon

This KITABU QUEST book turns the curriculum into short lessons, worked examples, guided activities, practice, review, and home links for Tanzania learners.

How To Use This Book

This book helps Form II learners study Biology one teachable topic at a time.

1. Start with the inquiry question and say what you already know.
2. Read the Learn section slowly and mark new words.
3. Try the worked example before reading the full solution.
4. Complete the activity and practice tasks.
5. Correct your answer using the success criteria.
6. Ask for support when your answer is still unclear.

Subject method: Use labelled diagrams, safe observations, evidence, and explanations about living things.

Learning Skills In This Book

Each topic builds knowledge, skill, values, communication, collaboration, critical thinking, creativity, digital awareness, and self-confidence.

Study actively: speak answers aloud, draw when useful, show your working, cite evidence, use local examples, and correct mistakes after feedback.

A strong answer is specific. It names the idea, gives evidence or method, applies it to the task, and checks whether it answers the question.

Table Of Contents

Use this table to move through the book one topic at a time. Each topic has a lesson opener, clear teaching, a model or example, guided work, practice, and checks for understanding.

1. Transport In Plants And Animals
2. Reproduction In Flowering Plants
3. Human Reproduction And Puberty
4. Growth Development And Nutrition
5. Respiration Photosynthesis And Energy Flow
6. Excretion And Homeostasis Basics
7. Ecology And Conservation
8. Form II Biology Project

As you finish each topic, return to the success criteria and correct one answer before moving on.

Transport In Plants And Animals: Lesson Opener

Learning focus: Transport In Plants And Animals

Country link: a safe Tanzania observation, model, data table, or investigation for Transport In Plants And Animals.

Progression focus: connect ideas across topics and explain causes, methods, or consequences; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:

- investigate and explain transport in plants and animals using evidence

Inquiry questions:

- How can transport in plants and animals help you solve a real problem?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Transport In Plants And Animals: Learn

Transport In Plants And Animals is learned by asking a clear question, observing carefully, recording evidence, and explaining what the evidence means.

Core idea:

- Start with the object, organism, material, body part, force, energy change, or environmental feature being studied.
- Use safe observations, labelled diagrams, simple measurements, or comparison tables.
- Keep evidence separate from guesses: evidence is what you saw, measured, read from a table, or labelled correctly.
- A conclusion should answer the question and mention one limit or next observation.
- Local examples from home, school, farms, water sources, weather, health, and the environment make the science useful.

Key words:

- Transport In Plants And Animals
- Transport
- Plants
- Animals
- concept
- observation
- evidence
- safety

Transport In Plants And Animals: Worked Example

Investigation model for Transport In Plants And Animals: Turn the topic into a safe, clear classroom investigation or model.

1. Question: write one question that can be answered by observing, comparing, classifying, measuring, modelling, or reading data.
2. Materials: list safe classroom materials, diagrams, samples, cards, tables, or teacher-approved apparatus.
3. Method: do one step at a time and keep one condition fair where comparison is needed.
4. Safety: name any heat, glass, sharp tool, chemical, hygiene, electrical, or body-safety rule before starting.
5. Record: use a labelled table, diagram, or sentence notes.
6. Conclusion: state what the evidence shows and one limit of the investigation.

Answer check: a strong response names the concept, uses correct vocabulary, records evidence, and does not claim more than the evidence supports.

Transport In Plants And Animals: Vocabulary And Study Skill

Use these words and study moves before you practise Transport In Plants And Animals. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Transport In Plants And Animals
- Transport
- Plants
- Animals
- concept
- observation
- evidence
- safety

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Tanzania.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Transport In Plants And Animals without a clear example, method, or evidence.

Transport In Plants And Animals: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Tanzania observation, model, data table, or investigation for Transport In Plants And Animals.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Transport In Plants And Animals: Practice And Correction

Practice for Transport In Plants And Animals:

Main outcome: investigate and explain transport in plants and animals using evidence.

1. Write one clear science question for the topic.
2. Use a labelled diagram, observation table, or model to record evidence.
3. Write a conclusion that answers the question and uses topic vocabulary.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Transport In Plants And Animals: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Transport In Plants And Animals.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Transport In Plants And Animals: Outcome Check

Outcome check for Transport In Plants And Animals: Investigate and explain transport in plants and animals using evidence.

1. State the science question.
2. Show the evidence with a labelled diagram, table, model, or observation.
3. Write a conclusion that follows from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Reproduction In Flowering Plants: Lesson Opener

Learning focus: Reproduction In Flowering Plants

Country link: a safe Tanzania observation, model, data table, or investigation for Reproduction In Flowering Plants.

Progression focus: connect ideas across topics and explain causes, methods, or consequences; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:

- investigate and explain reproduction in flowering plants using evidence

Inquiry questions:

- How can reproduction in flowering plants help you solve a real problem?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Reproduction In Flowering Plants: Learn

Reproduction In Flowering Plants teaches the human reproductive system with correct vocabulary, respect, privacy, and health awareness.

Core idea:

- Reproductive organs have specific structures and functions.
- Puberty brings physical and emotional changes that should be discussed respectfully.
- Personal hygiene, safety, consent, and seeking trusted adult or health-worker support are important.
- Diagrams should be labelled scientifically and handled with maturity.

Key words:

- Reproduction In Flowering Plants
- Reproduction
- Flowering
- Plants
- concept
- observation

Reproduction In Flowering Plants: Worked Example

Model for Reproduction In Flowering Plants: Label and explain a reproductive-system diagram respectfully.

1. Use the scientific diagram provided by the teacher or book.
2. Label the main structures with correct vocabulary.
3. Write one function for each labelled structure.
4. Add one health or hygiene point.
5. Keep discussion respectful and private.

Conclusion: correct labels and respectful explanation help learners understand body systems safely.

Reproduction In Flowering Plants: Vocabulary And Study Skill

Use these words and study moves before you practise Reproduction In Flowering Plants. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Reproduction In Flowering Plants
- Reproduction
- Flowering
- Plants
- concept
- observation
- evidence
- safety

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Tanzania.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Reproduction In Flowering Plants without a clear example, method, or evidence.

Reproduction In Flowering Plants: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Tanzania observation, model, data table, or investigation for Reproduction In Flowering Plants.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Reproduction In Flowering Plants: Practice And Correction

Practice for Reproduction In Flowering Plants:

Main outcome: investigate and explain reproduction in flowering plants using evidence.

1. Label a teacher-approved reproductive-system diagram.
2. Write one function for three labelled parts.
3. List two respectful health, hygiene, privacy, or safety points.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Reproduction In Flowering Plants: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Reproduction In Flowering Plants.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Reproduction In Flowering Plants: Outcome Check

Outcome check for Reproduction In Flowering Plants: Investigate and explain reproduction in flowering plants using evidence.

1. Label three structures on a teacher-approved diagram.
2. Write one function for each.
3. Add one respectful health or hygiene point.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Human Reproduction And Puberty: Lesson Opener

Learning focus: Human Reproduction And Puberty

Country link: a safe Tanzania observation, model, data table, or investigation for Human Reproduction And Puberty.

Progression focus: connect ideas across topics and explain causes, methods, or consequences; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:

- investigate and explain human reproduction and puberty using evidence

Inquiry questions:

- How can human reproduction and puberty help you solve a real problem?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Human Reproduction And Puberty: Learn

Human Reproduction And Puberty teaches the human reproductive system with correct vocabulary, respect, privacy, and health awareness.

Core idea:

- Reproductive organs have specific structures and functions.
- Puberty brings physical and emotional changes that should be discussed respectfully.
- Personal hygiene, safety, consent, and seeking trusted adult or health-worker support are important.
- Diagrams should be labelled scientifically and handled with maturity.

Key words:

- Human Reproduction And Puberty
- Human
- Reproduction
- Puberty
- concept
- observation
- evidence
- safety

Human Reproduction And Puberty: Worked Example

Model for Human Reproduction And Puberty: Label and explain a reproductive-system diagram respectfully.

1. Use the scientific diagram provided by the teacher or book.
2. Label the main structures with correct vocabulary.
3. Write one function for each labelled structure.
4. Add one health or hygiene point.
5. Keep discussion respectful and private.

Conclusion: correct labels and respectful explanation help learners understand body systems safely.

Human Reproduction And Puberty: Vocabulary And Study Skill

Use these words and study moves before you practise Human Reproduction And Puberty. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Human Reproduction And Puberty
- Human
- Reproduction
- Puberty
- concept
- observation
- evidence
- safety

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Tanzania.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Human Reproduction And Puberty without a clear example, method, or evidence.

Human Reproduction And Puberty: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Tanzania observation, model, data table, or investigation for Human Reproduction And Puberty.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Human Reproduction And Puberty: Practice And Correction

Practice for Human Reproduction And Puberty:

Main outcome: investigate and explain human reproduction and puberty using evidence.

1. Label a teacher-approved reproductive-system diagram.
2. Write one function for three labelled parts.
3. List two respectful health, hygiene, privacy, or safety points.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Human Reproduction And Puberty: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Human Reproduction And Puberty.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Human Reproduction And Puberty: Outcome Check

Outcome check for Human Reproduction And Puberty: Investigate and explain human reproduction and puberty using evidence.

1. Label three structures on a teacher-approved diagram.
2. Write one function for each.
3. Add one respectful health or hygiene point.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Growth Development And Nutrition: Lesson Opener

Learning focus: Growth Development And Nutrition

Country link: a safe Tanzania observation, model, data table, or investigation for Growth Development And Nutrition.

Progression focus: connect ideas across topics and explain causes, methods, or consequences; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:

- investigate and explain growth development and nutrition using evidence

Inquiry questions:

- How can growth development and nutrition help you solve a real problem?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Growth Development And Nutrition: Learn

Growth Development And Nutrition teaches useful computer work through exact steps, not guessing.

Core idea:

- Word processing is used to create, edit, format, save, and print text documents.
- A spreadsheet organizes data in rows, columns, and cells.
- A cell can hold text, a number, or a formula.
- Formatting should make work clearer, not just more decorative.
- Good digital work is saved with a clear file name and checked before sharing.

Learner task link: create a class list, simple budget, marks table, timetable, or short report using safe school devices.

Key words:

- Growth Development And Nutrition
- Growth
- Development
- Nutrition
- concept
- observation
- evidence
- safety

Growth Development And Nutrition: Worked Example

Worked example for Growth Development And Nutrition: Make a simple spreadsheet for class garden harvest.

1. Open a blank worksheet.
2. In row 1, type headings: Crop, Week 1, Week 2, Total.
3. Enter beans, maize, and tomatoes in the Crop column.
4. Type harvest numbers for Week 1 and Week 2.
5. In the Total column, use a formula such as =B2+C2.
6. Save the file with a clear name, for example garden-harvest-p6.

Quality check: headings are clear, numbers are in the correct cells, formulas give sensible totals, and the file is saved.

Growth Development And Nutrition: Vocabulary And Study Skill

Use these words and study moves before you practise Growth Development And Nutrition. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Growth Development And Nutrition
- Growth
- Development
- Nutrition
- concept
- observation
- evidence
- safety

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Tanzania.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Growth Development And Nutrition without a clear example, method, or evidence.

Growth Development And Nutrition: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Tanzania observation, model, data table, or investigation for Growth Development And Nutrition.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Growth Development And Nutrition: Practice And Correction

Practice for Growth Development And Nutrition:

Main outcome: investigate and explain growth development and nutrition using evidence.

1. Create a document or worksheet with a clear title and headings.
2. Enter five rows of useful class, garden, timetable, budget, or marks data.
3. Format the work neatly, save it with a clear file name, and explain one formula or formatting choice.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Growth Development And Nutrition: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Growth Development And Nutrition.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Growth Development And Nutrition: Outcome Check

Outcome check for Growth Development And Nutrition: Investigate and explain growth development and nutrition using evidence.

1. Create a two-column table or worksheet for a real class task.
2. Use one correct heading, formula, save step, or formatting step.
3. Explain how the digital file can be checked for errors.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Respiration Photosynthesis And Energy Flow: Lesson Opener

Learning focus: Respiration Photosynthesis And Energy Flow

Country link: a safe Tanzania observation, model, data table, or investigation for Respiration Photosynthesis And Energy Flow.

Progression focus: connect ideas across topics and explain causes, methods, or consequences; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:

- investigate and explain respiration photosynthesis and energy flow using evidence

Inquiry questions:

- How can respiration photosynthesis and energy flow help you solve a real problem?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Respiration Photosynthesis And Energy Flow: Learn

Respiration Photosynthesis And Energy Flow connects work, power, and energy changes.

Core idea:

- Work is done when a force moves an object through a distance.
- Energy is the ability to do work.
- Power describes how quickly work is done or energy is transferred.
- Energy can change form, such as chemical to movement, electrical to light, or potential to kinetic.

Key words:

- Respiration Photosynthesis And Energy Flow
- Respiration
- Photosynthesis
- Energy
- Flow
- concept
- observation
- evidence

Respiration Photosynthesis And Energy Flow: Worked Example

Worked example for Respiration Photosynthesis And Energy Flow: A learner lifts a book from the floor to a desk.

1. The learner applies an upward force.
2. The book moves upward through a distance.
3. Work is done on the book.
4. The book gains gravitational potential energy.

Conclusion: work can transfer energy to an object.

Respiration Photosynthesis And Energy Flow: Vocabulary And Study Skill

Use these words and study moves before you practise Respiration Photosynthesis And Energy Flow. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Respiration Photosynthesis And Energy Flow
- Respiration
- Photosynthesis
- Energy
- Flow
- concept
- observation
- evidence

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Tanzania.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Respiration Photosynthesis And Energy Flow without a clear example, method, or evidence.

Respiration Photosynthesis And Energy Flow: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Tanzania observation, model, data table, or investigation for Respiration Photosynthesis And Energy Flow.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Respiration Photosynthesis And Energy Flow: Practice And Correction

Practice for Respiration Photosynthesis And Energy Flow:

Main outcome: investigate and explain respiration photosynthesis and energy flow using evidence.

1. Give two examples of energy transformation.
2. Explain when work is done on an object.
3. Compare work and power in one sentence.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Respiration Photosynthesis And Energy Flow: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Respiration Photosynthesis And Energy Flow.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Respiration Photosynthesis And Energy Flow: Outcome Check

Outcome check for Respiration Photosynthesis And Energy Flow: Investigate and explain respiration photosynthesis and energy flow using evidence.

1. Give one example of energy changing form.
2. Explain when work is done.
3. Compare work and power in one sentence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Excretion And Homeostasis Basics: Lesson Opener

Learning focus: Excretion And Homeostasis Basics

Country link: a safe Tanzania observation, model, data table, or investigation for Excretion And Homeostasis Basics.

Progression focus: connect ideas across topics and explain causes, methods, or consequences; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:

- investigate and explain excretion and homeostasis basics using evidence

Inquiry questions:

- How can excretion and homeostasis basics help you solve a real problem?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Excretion And Homeostasis Basics: Learn

Excretion And Homeostasis Basics is learned by turning the topic into a clear, safe, evidence-based science task.

Core idea:

- Name the science idea before starting: structure, process, material, force, energy, organism, environment, or system.
- Use a diagram, data table, labelled model, safe observation, or teacher-approved investigation.
- Record what is seen or measured, not what is guessed.
- Explain the result using the vocabulary from this topic.
- Finish with one conclusion and one question for further study.

Key words:

- Excretion And Homeostasis Basics
- Excretion
- Homeostasis
- Basics
- concept
- observation
- evidence
- safety

Excretion And Homeostasis Basics: Worked Example

Investigation model for Excretion And Homeostasis Basics: Turn the topic into a safe, clear classroom investigation or model.

1. Question: write one question that can be answered by observing, comparing, classifying, measuring, modelling, or reading data.
2. Materials: list safe classroom materials, diagrams, samples, cards, tables, or teacher-approved apparatus.
3. Method: do one step at a time and keep one condition fair where comparison is needed.
4. Safety: name any heat, glass, sharp tool, chemical, hygiene, electrical, or body-safety rule before starting.
5. Record: use a labelled table, diagram, or sentence notes.
6. Conclusion: state what the evidence shows and one limit of the investigation.

Answer check: a strong response names the concept, uses correct vocabulary, records evidence, and does not claim more than the evidence supports.

Excretion And Homeostasis Basics: Vocabulary And Study Skill

Use these words and study moves before you practise Excretion And Homeostasis Basics. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Excretion And Homeostasis Basics
- Excretion
- Homeostasis
- Basics
- concept
- observation
- evidence
- safety

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Tanzania.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Excretion And Homeostasis Basics without a clear example, method, or evidence.

Excretion And Homeostasis Basics: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Tanzania observation, model, data table, or investigation for Excretion And Homeostasis Basics.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Excretion And Homeostasis Basics: Practice And Correction

Practice for Excretion And Homeostasis Basics:

Main outcome: investigate and explain excretion and homeostasis basics using evidence.

1. Write one safe investigation question.
2. Name materials and observations needed.
3. Use the evidence to write a conclusion.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Excretion And Homeostasis Basics: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Excretion And Homeostasis Basics.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Excretion And Homeostasis Basics: Outcome Check

Outcome check for Excretion And Homeostasis Basics: Investigate and explain excretion and homeostasis basics using evidence.

1. State a safe investigation question.
2. Name the evidence needed.
3. Write a conclusion from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Ecology And Conservation: Lesson Opener

Learning focus: Ecology And Conservation

Country link: a safe Tanzania observation, model, data table, or investigation for Ecology And Conservation.

Progression focus: connect ideas across topics and explain causes, methods, or consequences; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:

- investigate and explain ecology and conservation using evidence

Inquiry questions:

- How can ecology and conservation help you solve a real problem?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Ecology And Conservation: Learn

Ecology And Conservation is learned by turning the topic into a clear, safe, evidence-based science task.

Core idea:

- Name the science idea before starting: structure, process, material, force, energy, organism, environment, or system.
- Use a diagram, data table, labelled model, safe observation, or teacher-approved investigation.
- Record what is seen or measured, not what is guessed.
- Explain the result using the vocabulary from this topic.
- Finish with one conclusion and one question for further study.

Key words:

- Ecology And Conservation
- Ecology
- Conservation
- concept
- observation
- evidence
- safety
- conclusion

Ecology And Conservation: Worked Example

Investigation model for Ecology And Conservation: Turn the topic into a safe, clear classroom investigation or model.

1. Question: write one question that can be answered by observing, comparing, classifying, measuring, modelling, or reading data.
2. Materials: list safe classroom materials, diagrams, samples, cards, tables, or teacher-approved apparatus.
3. Method: do one step at a time and keep one condition fair where comparison is needed.
4. Safety: name any heat, glass, sharp tool, chemical, hygiene, electrical, or body-safety rule before starting.
5. Record: use a labelled table, diagram, or sentence notes.
6. Conclusion: state what the evidence shows and one limit of the investigation.

Answer check: a strong response names the concept, uses correct vocabulary, records evidence, and does not claim more than the evidence supports.

Ecology And Conservation: Vocabulary And Study Skill

Use these words and study moves before you practise Ecology And Conservation. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Ecology And Conservation
- Ecology
- Conservation
- concept
- observation

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Tanzania.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Ecology And Conservation without a clear example, method, or evidence.

Ecology And Conservation: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Tanzania observation, model, data table, or investigation for Ecology And Conservation.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Ecology And Conservation: Practice And Correction

Practice for Ecology And Conservation:

Main outcome: investigate and explain ecology and conservation using evidence.

1. Write one safe investigation question.
2. Name materials and observations needed.
3. Use the evidence to write a conclusion.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Ecology And Conservation: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Ecology And Conservation.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Ecology And Conservation: Outcome Check

Outcome check for Ecology And Conservation: Investigate and explain ecology and conservation using evidence.

1. State a safe investigation question.
2. Name the evidence needed.
3. Write a conclusion from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Form II Biology Project: Lesson Opener

Learning focus: Form II Biology Project

Country link: a safe Tanzania observation, model, data table, or investigation for Form II Biology Project.

Progression focus: connect ideas across topics and explain causes, methods, or consequences; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:

- investigate and explain form ii biology project using evidence

Inquiry questions:

- How can form ii biology project help you solve a real problem?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Form II Biology Project: Learn

Form II Biology Project is learned by turning the topic into a clear, safe, evidence-based science task.

Core idea:

- Name the science idea before starting: structure, process, material, force, energy, organism, environment, or system.
- Use a diagram, data table, labelled model, safe observation, or teacher-approved investigation.
- Record what is seen or measured, not what is guessed.
- Explain the result using the vocabulary from this topic.
- Finish with one conclusion and one question for further study.

Key words:

- Form II Biology Project
- Form
- Biology
- Project
- concept
- observation
- evidence
- safety

Form II Biology Project: Worked Example

Investigation model for Form II Biology Project: Turn the topic into a safe, clear classroom investigation or model.

1. Question: write one question that can be answered by observing, comparing, classifying, measuring, modelling, or reading data.
2. Materials: list safe classroom materials, diagrams, samples, cards, tables, or teacher-approved apparatus.
3. Method: do one step at a time and keep one condition fair where comparison is needed.
4. Safety: name any heat, glass, sharp tool, chemical, hygiene, electrical, or body-safety rule before starting.
5. Record: use a labelled table, diagram, or sentence notes.
6. Conclusion: state what the evidence shows and one limit of the investigation.

Answer check: a strong response names the concept, uses correct vocabulary, records evidence, and does not claim more than the evidence supports.

Form II Biology Project: Vocabulary And Study Skill

Use these words and study moves before you practise Form II Biology Project. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Form II Biology Project
- Form
- Biology
- Project
- concept
- observation

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Tanzania.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Form II Biology Project without a clear example, method, or evidence.

Form II Biology Project: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Tanzania observation, model, data table, or investigation for Form II Biology Project.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Form II Biology Project: Practice And Correction

Practice for Form II Biology Project:

Main outcome: investigate and explain form ii biology project using evidence.

1. Write one safe investigation question.
2. Name materials and observations needed.
3. Use the evidence to write a conclusion.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Form II Biology Project: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Form II Biology Project.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Form II Biology Project: Outcome Check

Outcome check for Form II Biology Project: Investigate and explain form ii biology project using evidence.

1. State a safe investigation question.
2. Name the evidence needed.
3. Write a conclusion from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Transport In Plants And Animals: Review Clinic 1

Use this review clinic to strengthen Biology without copying earlier answers.

1. Choose one idea from Transport In Plants And Animals that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Tanzania.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Reproduction In Flowering Plants: Review Clinic 2

Use this review clinic to strengthen Biology without copying earlier answers.

1. Choose one idea from Reproduction In Flowering Plants that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Tanzania.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Human Reproduction And Puberty: Review Clinic 3

Use this review clinic to strengthen Biology without copying earlier answers.

1. Choose one idea from Human Reproduction And Puberty that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Tanzania.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Growth Development And Nutrition: Review Clinic 4

Use this review clinic to strengthen Biology without copying earlier answers.

1. Choose one idea from Growth Development And Nutrition that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Tanzania.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Transport In Plants And Animals: Application Workshop 1

Application workshop 1 for Transport In Plants And Animals.

Grade move: connect ideas across topics and explain causes, methods, or consequences; connect evidence, diagrams, safety, and conclusions.

Use this page to deepen the science idea with evidence.

1. Write one question that can be answered by observation, model, measurement, classification, or data reading.
2. Name safe materials, diagrams, samples, tables, or apparatus that would help.
3. Predict what you expect and explain why.
4. Record evidence in a labelled table, diagram, or sentence notes.
5. Write a conclusion that answers the question without adding unsupported guesses.
6. Add one safety rule and one limitation of the evidence.

Extension: suggest one improved investigation for a future lesson.

Reproduction In Flowering Plants: Application Workshop 2

Application workshop 2 for Reproduction In Flowering Plants.

Grade move: connect ideas across topics and explain causes, methods, or consequences; connect evidence, diagrams, safety, and conclusions.

Use this page to deepen the science idea with evidence.

1. Write one question that can be answered by observation, model, measurement, classification, or data reading.
2. Name safe materials, diagrams, samples, tables, or apparatus that would help.
3. Predict what you expect and explain why.
4. Record evidence in a labelled table, diagram, or sentence notes.
5. Write a conclusion that answers the question without adding unsupported guesses.
6. Add one safety rule and one limitation of the evidence.

Extension: suggest one improved investigation for a future lesson.

Human Reproduction And Puberty: Application Workshop 3

Application workshop 3 for Human Reproduction And Puberty.

Grade move: connect ideas across topics and explain causes, methods, or consequences; connect evidence, diagrams, safety, and conclusions.

Use this page to deepen the science idea with evidence.

1. Write one question that can be answered by observation, model, measurement, classification, or data reading.
2. Name safe materials, diagrams, samples, tables, or apparatus that would help.
3. Predict what you expect and explain why.
4. Record evidence in a labelled table, diagram, or sentence notes.
5. Write a conclusion that answers the question without adding unsupported guesses.
6. Add one safety rule and one limitation of the evidence.

Extension: suggest one improved investigation for a future lesson.

Growth Development And Nutrition: Application Workshop 4

Application workshop 4 for Growth Development And Nutrition.

Grade move: connect ideas across topics and explain causes, methods, or consequences; connect evidence, diagrams, safety, and conclusions.

Use this page to deepen the science idea with evidence.

1. Write one question that can be answered by observation, model, measurement, classification, or data reading.
2. Name safe materials, diagrams, samples, tables, or apparatus that would help.
3. Predict what you expect and explain why.
4. Record evidence in a labelled table, diagram, or sentence notes.
5. Write a conclusion that answers the question without adding unsupported guesses.
6. Add one safety rule and one limitation of the evidence.

Extension: suggest one improved investigation for a future lesson.

Respiration Photosynthesis And Energy Flow: Application Workshop 5

Application workshop 5 for Respiration Photosynthesis And Energy Flow.

Grade move: connect ideas across topics and explain causes, methods, or consequences; connect evidence, diagrams, safety, and conclusions.

Use this page to deepen the science idea with evidence.

1. Write one question that can be answered by observation, model, measurement, classification, or data reading.
2. Name safe materials, diagrams, samples, tables, or apparatus that would help.
3. Predict what you expect and explain why.
4. Record evidence in a labelled table, diagram, or sentence notes.
5. Write a conclusion that answers the question without adding unsupported guesses.
6. Add one safety rule and one limitation of the evidence.

Extension: suggest one improved investigation for a future lesson.

Excretion And Homeostasis Basics: Application Workshop 6

Application workshop 6 for Excretion And Homeostasis Basics.

Grade move: connect ideas across topics and explain causes, methods, or consequences; connect evidence, diagrams, safety, and conclusions.

Use this page to deepen the science idea with evidence.

1. Write one question that can be answered by observation, model, measurement, classification, or data reading.
2. Name safe materials, diagrams, samples, tables, or apparatus that would help.
3. Predict what you expect and explain why.
4. Record evidence in a labelled table, diagram, or sentence notes.
5. Write a conclusion that answers the question without adding unsupported guesses.
6. Add one safety rule and one limitation of the evidence.

Extension: suggest one improved investigation for a future lesson.

Ecology And Conservation: Application Workshop 7

Application workshop 7 for Ecology And Conservation.

Grade move: connect ideas across topics and explain causes, methods, or consequences; connect evidence, diagrams, safety, and conclusions.

Use this page to deepen the science idea with evidence.

1. Write one question that can be answered by observation, model, measurement, classification, or data reading.
2. Name safe materials, diagrams, samples, tables, or apparatus that would help.
3. Predict what you expect and explain why.
4. Record evidence in a labelled table, diagram, or sentence notes.
5. Write a conclusion that answers the question without adding unsupported guesses.
6. Add one safety rule and one limitation of the evidence.

Extension: suggest one improved investigation for a future lesson.

Form II Biology Project: Application Workshop 8

Application workshop 8 for Form II Biology Project.

Grade move: connect ideas across topics and explain causes, methods, or consequences; connect evidence, diagrams, safety, and conclusions.

Use this page to deepen the science idea with evidence.

1. Write one question that can be answered by observation, model, measurement, classification, or data reading.
2. Name safe materials, diagrams, samples, tables, or apparatus that would help.
3. Predict what you expect and explain why.
4. Record evidence in a labelled table, diagram, or sentence notes.
5. Write a conclusion that answers the question without adding unsupported guesses.
6. Add one safety rule and one limitation of the evidence.

Extension: suggest one improved investigation for a future lesson.

Transport In Plants And Animals: Application Workshop 9

Application workshop 9 for Transport In Plants And Animals.

Grade move: connect ideas across topics and explain causes, methods, or consequences; connect evidence, diagrams, safety, and conclusions.

Use this page to deepen the science idea with evidence.

1. Write one question that can be answered by observation, model, measurement, classification, or data reading.
2. Name safe materials, diagrams, samples, tables, or apparatus that would help.
3. Predict what you expect and explain why.
4. Record evidence in a labelled table, diagram, or sentence notes.
5. Write a conclusion that answers the question without adding unsupported guesses.
6. Add one safety rule and one limitation of the evidence.

Extension: suggest one improved investigation for a future lesson.

Reproduction In Flowering Plants: Application Workshop 10

Application workshop 10 for Reproduction In Flowering Plants.

Grade move: connect ideas across topics and explain causes, methods, or consequences; connect evidence, diagrams, safety, and conclusions.

Use this page to deepen the science idea with evidence.

1. Write one question that can be answered by observation, model, measurement, classification, or data reading.
2. Name safe materials, diagrams, samples, tables, or apparatus that would help.
3. Predict what you expect and explain why.
4. Record evidence in a labelled table, diagram, or sentence notes.
5. Write a conclusion that answers the question without adding unsupported guesses.
6. Add one safety rule and one limitation of the evidence.

Extension: suggest one improved investigation for a future lesson.

Glossary

Use this glossary to revise important words in Biology.

- Transport In Plants And Animals: Explain this term using an example from the book, your school, or your community.
- Reproduction In Flowering Plants: Explain this term using an example from the book, your school, or your community.
- Human Reproduction And Puberty: Explain this term using an example from the book, your school, or your community.
- Growth Development And Nutrition: Explain this term using an example from the book, your school, or your community.
- Respiration Photosynthesis And Energy Flow: Explain this term using an example from the book, your school, or your community.
- Excretion And Homeostasis Basics: Explain this term using an example from the book, your school, or your community.
- Ecology And Conservation: Explain this term using an example from the book, your school, or your community.
- Form II Biology Project: Explain this term using an example from the book, your school, or your community.

Add five more words from your lessons and write your own meanings.

Answer And Teacher Notes

A strong Biology answer states the idea, uses correct scientific vocabulary, records observations or data, includes units or labels where needed, and writes a conclusion that follows from evidence. Safety rules always come first. If evidence is missing, repeat the observation safely or explain what evidence would be needed.

Final Project

Create a final project for Biology that shows what you can now do independently.

Your project must include:

1. A clear title.
2. A real-life problem, question, text, performance, investigation, design, or worked task.
3. Evidence from at least three lessons in this book.
4. A drawing, table, map, chart, model, dialogue, paragraph, performance plan, practical product, or worked solution.
5. A correction note showing one improvement after feedback.
6. A short reflection explaining what you learned, what was difficult, and what you will practise next.

Presentation check: keep the work neat, label important parts, use correct vocabulary, and be ready to explain your choices to a classmate or teacher.